

Bangabandhu Sheikh Mujibur Rahman Science and Technology University

Department of Computer Science and Engineering

3rd Year 2nd Semester B.Sc. Engineering Examination - 2021

Course Code: CSE357

Course Title: Microprocessor and Microcontroller

Total Marks: 60

Time: 3 (Three) Hours

N.B.

- i. Answer **SIX** questions taking any **THREE** from each section.
- ii. All parts of a question must be answered sequentially.

Section-A

1. a) Write a macro that computes factorial of a number in assembly code. 4
b) What does addressing mode mean? Discuss the several addressing modes used by the 8086 microprocessor. 3
c) Explain the internal architecture of the 8086 microprocessor using a block diagram 3
2. a) What do you mean by 8-bit and 16-bit microprocessor? 3
b) Describe the general purpose registers 4
c) What happens when the microprocessor is powered up? 3
3. a) What is the Flag register? Discuss the role of five flag registers in an ALU operation using an example. 4
b) Draw the block diagram of 8086 microcomputer system overview and Explain how data is transferred with in the 8086 microcomputer system. 3
c) What is memory segmentation? How different memory segments are accessed in 8086 microprocessor? 3
4. a) Distinguish RISC, CISC and special microprocessor. 3
b) What do you mean by program segment prefix? 1
c) Distinguish address, data and control bus. 3
d) Write an assembly language program to add two decimal numbers. 3

Section-B

5. a) What is the purpose of the FLAGS register? 3.5
b) Distinguishes between Logical address and Physical address. 3
c) Write short notes on following terms: 3.5
i) LEA
ii) MOY
6. a) Depicts the Intel 8086 Microprocessor Organization. 4
b) What are the differences between microprocessor and microcontroller? 3
c) Why do we need latch in 8086 microprocessor? 3
7. a) Sub AX, BX, where AX contains 8000h and BX contains 0001h and calculate Status flag, Parity flag, Zero flag, Carry flag and Overflow flag. 3
b) Draw the block diagram of 8051 microprocessor and describe the components of the hardware. 4
c) What do you mean by .Stack 100h in assembly language? 3
8. a) Why flag is used in microprocessor? Differentiate between status and control flag. 3
b) Write short notes on following terms: 5
i. LEA
ii. MOV
iii. PUSH
iv. POP
c) What do you mean by .Stack 100h in assembly language? 2

Bangabandhu Sheikh Mujibur Rahman Science and Technology University
Department of Computer Science and Engineering (ICT)
3rd Year 2nd Semester B.Sc. Engineering Examination-2021

Course Code: CSE359

Course Title: Software Engineering

Total Marks: 60

Time: 3 (Three) Hours

N.B.:

- i) Answer SIX questions taking any THREE from each section.
- ii) All parts of a question must be answered sequentially.

Section-A

1. a) What is the aim of software engineering? Illustrate with a diagram that the software does not wear out. 1+3
b) How are software myths affecting software process? Explain with the help of examples. 3
c) What do you understand by the term SDLC? Why is it important to adhere to a life cycle model while developing a large software product? 1+2
2. a) Describe each of the following terms: 4
 i. JRP
 ii. System testing
b) Describe factory design pattern with proper examples. 4
c) What are the attributes of a good software? 2
3. a) What are fact finding methods? As a part of ensuring eGovernance goals, the Government is going to computerize its land management process that keeps records about land ownership and transfers of ownerships. It is your responsibility to do the requirement analysis for such a project. What fact finding techniques you may use and how? 4
b) A book is written by an author, published by a publisher, sold by a bookstore, and read by a reader. Moreover, for a reader to read a book, she must buy it from a book store that is selling it. 4
 Draw a Use Case diagram for this scenario, showing relationships between different use cases.
c) What are the different types of messages in a sequence diagram? 2
4. a) To buy a book electronically from chapters.com, a customer needs to select the book from a list provided by Chapter's eCommerce system, provide credit card information to the system, then the system gets authorization from the bank for the payment, and --if positive -- confirms the sale. The order is then sent to the orders department and when the book becomes available, it is shipped to the customer. Also, the order department charges the customer's credit card by informing the bank of the amount. 5
 Draw a Sequence diagram that models this process. Make sure to model all relevant actors and the interactions between them. Do show explicitly the time intervals when different actors actively participate in the process you are modeling.
b) Why is software testing important? Who does it? What are the attributes of a good test? 5

Section-B

5. a) What is the difference between verification and validation? What input and output faults you may check while designing a banking transaction module. 4
b) Represent a hospital management system using layered architectural and client server architectural patterns with necessary justifications. 4
c) Describe the agile manifesto. 2

6. a) Consider, you have a collection of several hundred music CDs. Your friends love to borrow them, often borrowing several at the same time. You don't mind lending the CDs, but it is becoming difficult for you to keep track of which friend borrowed which CD. The difficulty is increased by the fact that you have more than one copy of some CDs. Also, you are thinking of introducing a daily fee per CD loaned. For these reasons, you would like to build a database, which you could query to find out how many CDs a given friend has; how long the friend has had them and how much he or she owes; who has a particular CD and so on. Draw a class **diagram** for the above scenario. 5
- b) Show example diagrams of real-world scenarios to present Aggregation and Composition in the context of class diagrams. 3
- c) How do you define software and software engineering? 2
7. a) What do you know about COCOMO model? Describe the basic COCOMO model with its limitations. 4
- b) Briefly explain the purpose of each of the sections in a software project plan. 4
- c) What is the difference between a software process model and a software process? 2
8. a) What is risk identification? Suppose you are a project manager of 'ETC software firm', assume that high staff turnover is noted as project risk in your project team. Now which steps should you take to mitigate this risk? 4
- b) Which of the following (i) and (ii) is a valid implementation of **Singleton Pattern**? Give specific reasons for each of the following implementations. 3
- (i)
- ```
public class Singleton {
 private static Singleton s_singleton = new Singleton ();
 private Singleton () {}
 private static Singleton getInstance() {
 return s_singleton;
 }
}
```
- (ii)
- ```
public class Singleton {
    private Singleton () {}
    private static class SingletonHolder {
        private static final Singleton instance = new Singleton ();
    }
    public static Singleton getInstance () {
        return SingletonHolder. instance;
    }
}
```
- c) Describe the following software cost estimation techniques: 3
- Parkinson's Law.
 - Pricing to win.

N.B.:

- i) Answer **SIX** questions taking any **THREE** from each section.
- ii) All parts of a question must be answered sequentially.

Section-A

1. a) Why do need to study Numerical Methods? 2
 b) Explain: 4
 - i) Accuracy and precision.
 - ii) True error and approximation error.
- c) What is truncation error? How can we use Taylor series to estimate truncation error? Give an example. 4
2. a) If $f(x)$ is real and continuous in the interval x_1 to x_2 , then what should be the condition that there can be at least one real root between x_1 and x_2 to this function? Explain the reason. 2
 b) Give stepwise explanation of Bisection method. 3
 c) What are differences between secant method and false-position method? 3
 d) Discuss the pitfall of false-position method and its solution. 2
3. a) Determine the lowest positive root of $f(x) = 7\sin(x) e^{-x} - 1$ using the Newton-Raphson method (three iterations, $x_1 = 0.3$). 5
 b) Perform the LU decomposition on the following equations: 5

$$\begin{aligned} 3x_1 - 0.1x_2 - 0.2x_3 &= 7.85 \\ 0.1x_1 + 7x_2 - 0.3x_3 &= -19.3 \\ 0.3x_1 - 0.2x_2 + 10x_3 &= 71.4 \end{aligned}$$
4. a) Determine a root of $f(x) = x^3 - 6x^2 + 11x - 6.1$ using modified secant method with an initial guess of $x = 3.5$ and $\delta = 0.001$. 5
 b) The upward velocity of a rocket is given at three different times. 5

Time, t (s)	Velocity, v (ms^{-1})
5	50
8	65
12	90

The velocity data is approximated by a polynomial as: $v(t) = a_1 t^2 + a_2 t + a_3$
 Find the velocity at $t = 6, 9$ and 11 seconds.

Section-B

5. a) Use Gauss elimination with partial pivoting to solve for the x 's : 6

$$\begin{aligned} -3x_2 + 7x_3 &= 2 \\ x_1 + 2x_2 - x_3 &= 3 \\ 5x_1 - 2x_2 &= 2 \end{aligned}$$
- b) Mention two pitfalls of Gauss Elimination method and the solution for those pitfalls. 4
6. a) Fit a straight line to the provided data using the least-squares regression principle. 6

x	1	2	3	4	5	6	7
y	0.5	2.5	2.0	4.0	3.5	6.0	5.5
- b) Solve the following equations with Gauss-Seidel method to $\epsilon_r = 5\%$.

$$\begin{aligned} 2x_1 - 6x_2 - x_3 &= -38 \\ -3x_1 - x_2 + 7x_3 &= -34 \\ -8x_1 + x_2 - 2x_3 &= -20 \end{aligned}$$

$a = -0.8$
 $b = 0.91$

7. a) Fit a third-order Newton's divided-difference interpolating polynomial to estimate $\ln(2)$ using the data:

$$x_0 = 1, \ln(x_0) = 0$$

$$x_1 = 4, \ln(x_1) = 1.386294$$

$$x_2 = 5, \ln(x_2) = 1.609438$$

$$x_3 = 6, \ln(x_3) = 1.791759$$

Compute the true percent relative error. Note that the true value of $\ln(2) = 0.6931472$.

- b) Give examples of different interpolating polynomials with proper examples.

8. a) Integrate the following function using Simpson's rules, with $n = 5$.

$$\int_{-3}^5 (4x - 3)^3 dx$$

- b) Find the area of a circle center at $(4, 5)$ and radius 3.4 using trapezoidal rules.

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Course Code: CSE351

Total Marks: 60

N.B.:

- i) Answer SIX questions taking any THREE from each section.
- ii) All parts of a question must be answered sequentially.

Course Title: Computer Graphics

Time: 3 (Three) Hours

Section-A

1. a) Define Computer graphics. What is pixel? 3
b) Explain the DDA algorithm for scan converting a line. How it take advantages against to direct use of line equation? 5
c) Why layering is needed in computer networking? 2
2. a) Digitize a line from (10, 12) to (15, 15) on a raster screen using mid-point line drawing algorithm. 3
b) Calculate the pixel location approximating the first octant of a circle having center at (4, 5) and radius 4 units using Bresenham's algorithm. 5
c) What is Transformation? 2
3. a) Explain in detail about two dimensional viewing. 4
b) Define Rotation. Explain the following basic two dimensional geometric transformations – (i) Translation (ii) Rotation. 4
c) Define Scaling and what are the types of scaling. 2
4. a) What is Composite transformation? Explain the following composite transformations (i) Translation (ii) Rotation 4
b) Define Reflection. What are the types of Clipping? 3
c) Define Shear. Explain it with example. 3

Section-B

5. a) Define Window. Define view port. 3
b) Differentiate parallel projection from perspective projection. 3
c) What is cubic spline? Give example of cubic spline. 4
6. a) What is Transformation? What are the steps involved in 3D transformation? 3
b) Write short notes on Bezier curve and spline. 4
c) Find a transformation of triangle A(1, 0), B(0, 1), C(1, 1) by 3
i. Rotating 45° about the origin and then translating one unit in x and y direction.
ii. Translating one unit in x and y direction and then rotating 45° about the origin.
7. a) Compare between geometric and coordinate transformation in 3D space. 3
b) "To project an image, engineers and drafters use parallel projection."- justify this. 3
c) Distinguish between orthographic and oblique parallel projection. 4
8. a) Define fractal. What is a Fractal Dimension? Give an example of fractals. 3
b) Write application of fractal in computer graphics. 3
c) What are the hardware devices used for computer graphics? What are the raster and vector graphics? 4

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Course No.: CSE353
 Full Marks: 60

Course Title: Computer Networks
 Time: 3 Hours

[Answer any Six questions taking Three from each section]

Section A

- Q.1 (a) Describe OSI Reference Model. 5
 (b) What are the differences between Public and Private Address? 2
 (c) How TCP connection Establish and close? 3
- Q.2 (a) Write short note on i. DNS ii. SMTP 4
 (b) Find out the hamming code for "CSE". 6
- Q.3 (a) What will be the transmitted frame if the given frame is 101100111 using the generator: 4
 $G(x) = x^3 + x^2 + 1$ when CRC code is used?
 (b) Explain how find ip address. 3
 (c) Mention the features of TCP and UDP. 3
- Q.4 (a) Write down the differences between IPV4 and IPV6. 2
 (b) Suppose In your university has a range of IP addresses 192.168.68.0/22. You have to create 8
 at least 7 usable subnets so that each subnet contains as many addresses as possible. Answer
 the following:
 i. What is the class of given IP block?
 ii. How many usable subnets will be created?
 iii. How many usable IP addresses will be there in each subnet?
 iv. What will be the subnet mask of the fourth subnet?
 v. What will be the second usable IP address of the second usable subnet?
 vi. Find out how many IPs are being lost in this subnetting.

Section B

- Q.5 (a) Describe the two-dimensional parity check error detection methods with example. 4
 (b) Describe the RSA algorithm for the message "BSMRSTU". 6
- Q.6 (a) How Collision handled with the CSMA/CD? 4
 (b) Discuss the relationship between frame and packet. 2
 (c) Find out the cipher text of "Bangabandhu Sheikh Mujibur Rahman" using Caesar cipher and 4
 transposition cipher method (consider 5 column).
- Q.7 (a) Briefly explain ARP and RARP address mapping technique. 6
 (b) What is the internet protocol? 2
 (c) What are port addresses and socket addresses? 2
- Q.8 (a) What is a routing table? Explain with an example. 4
 (b) What is DNS? Explain the purpose of DNS on the internet. 4
 (c) Discuss the basic model of FTP. 2

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